WORKSHEET

Chapter 1: Vector product

1 Calculate the vector product for each situation below:

a $|\mathbf{p}| = 6$, $|\mathbf{q}| = 5$, angle between $\mathbf{p}$ and $\mathbf{q} = 36°$

b $|\mathbf{a}| = 12$, $|\mathbf{b}| = 7$, angle between $\mathbf{a}$ and $\mathbf{b} = 121°$

c $|\mathbf{m}| = 3$, $|\mathbf{n}| = 9$, angle between $\mathbf{m}$ and $\mathbf{n} = 15°$

d $|\mathbf{r}| = 11$, $|\mathbf{t}| = 8$, angle between $\mathbf{r}$ and $\mathbf{t} = 87°$

e $|\mathbf{u}| = 13$, $|\mathbf{v}| = 21$, angle between $\mathbf{u}$ and $\mathbf{v} = 45°$

f $|\mathbf{d}| = 18$, $|\mathbf{e}| = 10$, angle between $\mathbf{d}$ and $\mathbf{e} = 155°$

2 Given the vectors $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$, $\mathbf{c} = 3\mathbf{i} - 5\mathbf{j} - 2\mathbf{k}$, $\mathbf{d} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{e} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, find:

a $\mathbf{a} \times \mathbf{b}$

b $\mathbf{c} \times \mathbf{e}$

c $\mathbf{e} \times \mathbf{d}$

d $\mathbf{c} \times (\mathbf{d} - \mathbf{e})$

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e $\mathbf{c} \times \mathbf{d} - \mathbf{c} \times \mathbf{e}$

f $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

g $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$

3 If $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ and $\mathbf{c} = -2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$, show that:
$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} \neq \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

4 For the vectors $\mathbf{m} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{n} = -\mathbf{i} - \mathbf{j} - \mathbf{k}$ show that, in this special case, $\mathbf{m} \times \mathbf{n} = \mathbf{n} \times \mathbf{m}$.

5 For the vectors $\mathbf{p} = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$, $\mathbf{q} = 5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$, $\mathrm{r} = \mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$, show that
$\mathbf{p} \times (\mathbf{q} + \mathbf{r}) = (\mathbf{p} \times \mathbf{q}) + (\mathbf{p} \times \mathbf{r})$

6 Use the vector product to find the areas of triangles with vertices:

a $(1, 2, -3)$, $(1, 5, 0)$, and $(5, 6, -1)$

b $(1, 1, 1)$, $(1, -1, 2)$, and $(0, 2, 3)$

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c $(0, 1, 4)$, $(-5, 9, 2)$, and $(7, 2, 8)$

7 Use the vector product to find the area of the parallelogram with vertices $P(1, -1, 4)$, $Q(4, 0, 3)$, $R(2, 2, -2)$ and $S(4, 5, 2)$.

8 Find the area of the parallelogram with adjacent sides given by the vectors in each case below.

a $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $7\mathbf{i} + \mathbf{j} + \mathbf{k}$

b $-\mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$ and $2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

c $7\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\mathbf{i} + 6\mathbf{j}$

9 If $PQRS$ is a parallelogram with $P(2, 1, 4)$, $Q(1, 4, 3)$ and $R(1, 0, 2)$, find:

a the area of the parallelogram

b the coordinates of S

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Answers

1 **a** Approximately 17.6 in a direction perpendicular to the plane of **p** and **q**.

b Approximately 84.9 in a direction perpendicular to the plane of **a** and **b**.

c Approximately 7 in a direction perpendicular to the plane of **m** and **n**.

d Approximately 87.9 in a direction perpendicular to the plane of **r** and **t**.

e Approximately 193 in a direction perpendicular to the plane of **u** and **v**.

f Approximately 76.1 in a direction perpendicular to the plane of **d** and **e**.

2 **a** $-\mathbf{i} + 13\mathbf{j} + 5\mathbf{k}$

b $-26\mathbf{i} - 16\mathbf{j} + \mathbf{k}$

c $-\mathbf{i} - 2\mathbf{j} - \mathbf{k}$

d $7\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$

e $7\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$

f $50\mathbf{i} - 65\mathbf{j} - 55\mathbf{k}$

g $-\mathbf{i} + 13\mathbf{j} - 34\mathbf{k}$

3–5 Demonstrations

6 **a** 9

b 2.74

c 35

7 16.16

8 **a** 21.4

b 28.6

c 49.5

9 **a** 8.12

b (2, −3, 3)

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